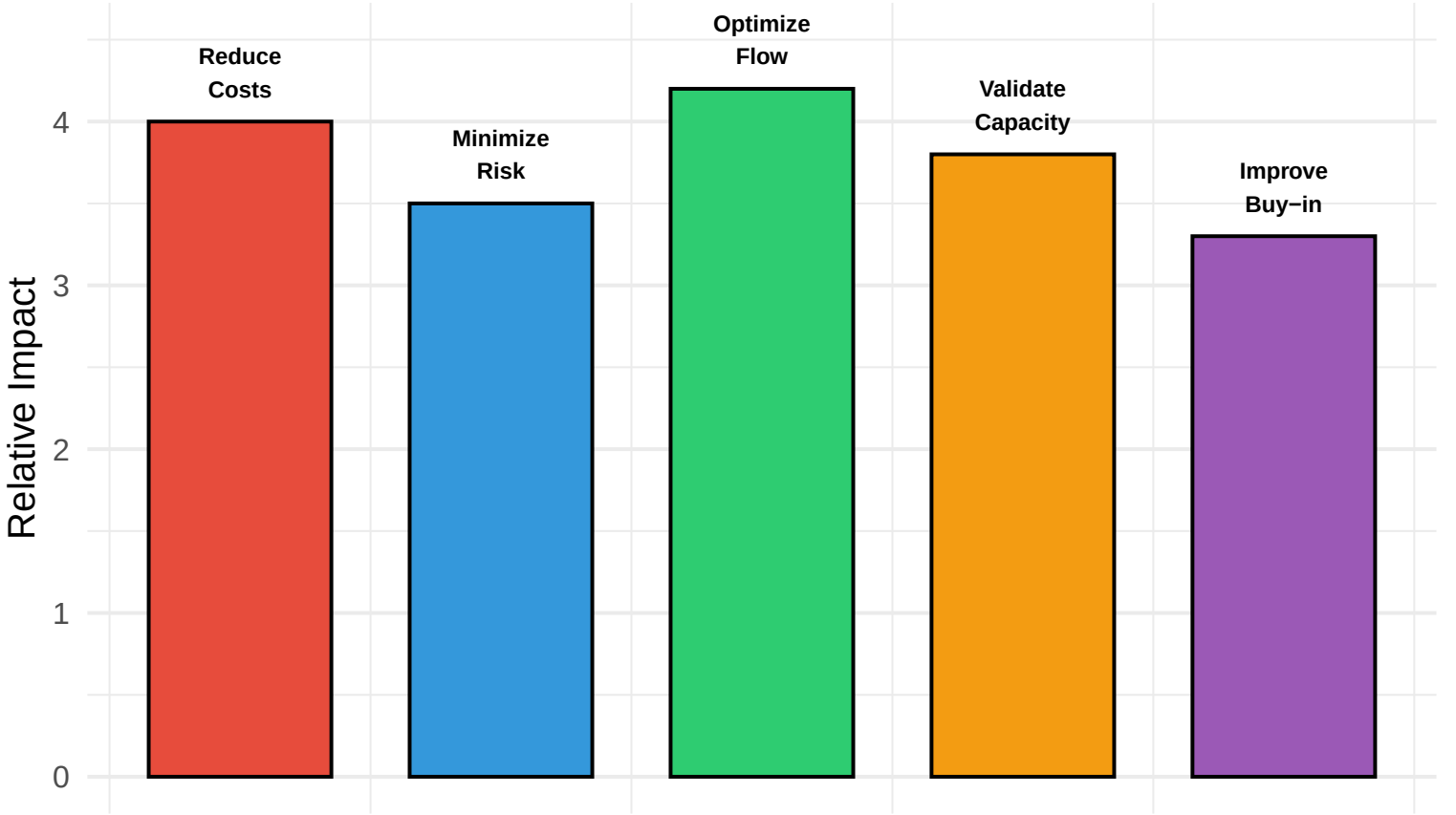Why Simulate Before Building?